KARAN KUMAR

+91 7905926475 karangupta11006@gmail.com karan-kumar0110 karankumar-hub

SUMMARY

To begin my professional journey in a dynamic organization where I can utilize my foundation in Electronics and Communication Engineering to contribute to innovative technological solutions, while enhancing my technical expertise, analytical abilities, and professional growth through continuous learning.

EDUCATION

- **Raj Kumar Goel Institute of Technology, Ghaziabad** 2022-26
  Bachelor of Technology in Electronics and Communication Engineering
- **Central Public Academy, Gorakhpur - Senior Secondary** 2022
  Central Board of Secondary Education (CBSE)
- **Renaissance Academy, Gorakhpur - High School** 2020
  Central Board of Secondary Education (CBSE)

EXPERIENCE

VLSI Training at VLSI Insights (Hybrid Mode)

- Digital Logic Design, CMOS Fundamentals, and Verilog/VHDL basics
- Simulation & Functional Verification using tools like Model Sim and Xilinx ISE
- Synthesis, Timing Analysis, and FPGA Implementation

ACADEMIC PROJECT

Autonomous Vacuum Cleaner

- Fully autonomous operation with obstacle avoidance
- Equipped with ultrasonic sensor for navigation and collision detection
- Vacuum suction unit to clean dust and small debris

Ultrasonic Radar System

- Real-time obstacle detection within 2–100 cm range
- 180° scanning using servo motor with radar-like GUI display
- Distance and angle data display for each detected object

RF Controlled Bot

- Wireless control using RF transmitter and receiver
- Directional movement using DC motors and a motor driver (L293D)
- Microcontroller (Arduino/Atmega328P) to decode the signals and control the bot

SKILLS

Programming & Tools: Python, Verilog, MATLAB, OrCAD PSpice, Arduino IDE, Processing IDE
Technical: Microcontroller, Digital Electronics, Embedded System, Analog & Digital Circuits, Op-Amps
Soft Skills: Analytical Thinking, Self-learning, Adaptability, Problem Solving, Teamwork & Collaboration
Relevant Coursework: Electronic Devices, Analog Circuits, Communication Engineering, Digital Electronics
PCB Designing: PCB Design Basics**,** Schematic Design & Component Placement, Routing Basics, Tool-EasyEDA

CERTIFICATE

VLSI Training (VLSI Insights)
Digital Design with Verilog (NPTEL)
Python Programming (INTERNSHALA)

POSITIONS & ACTIVITIES

Member – Organizing Committee (College Fest)
Member – Discipline Committee (Cultural Events)
Volunteer – IEEE Conference (Technical Assistance)
Volunteer – Vidyarthi Vigyan Manthan (STEM Outreach)